ZIM School - Use Cases

05. Attendance: daily roll calls and the monthly SF2

Modules: Attendance
3 use cases (changes) · 4 views and reports
Written from the use case commands, handlers, policies and rules in `src/` on 2026-11-10.

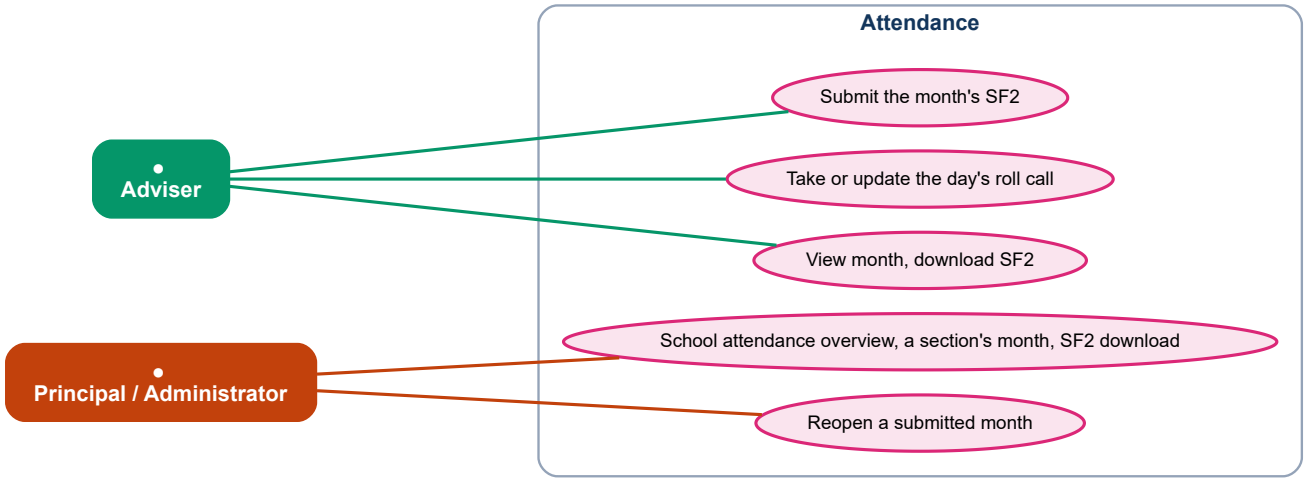
1. What this area does

The section's **adviser** takes a roll call for each school day. Only exceptions are stored: a learner who is absent, late or cutting classes gets a mark; everyone else counts as present. At the end of the month the adviser **submits the SF2** (School Form 2, Daily Attendance Report) to the principal, which locks the month. The principal (or an administrator) can **reopen** a submitted month so the adviser can correct it.

Who counts on a day comes from the Learners API (`SectionRosterInterface`): learners who joined later or left by transfer only count on their days. School days come from the Schools calendar (weekdays inside the school year that are not days off). Every change takes the `attendance-section-<id>` lock. Who may do what is `AttendancePolicy`: the adviser takes and submits (current year only); the principal and administrators view and reopen.

2. Use case map

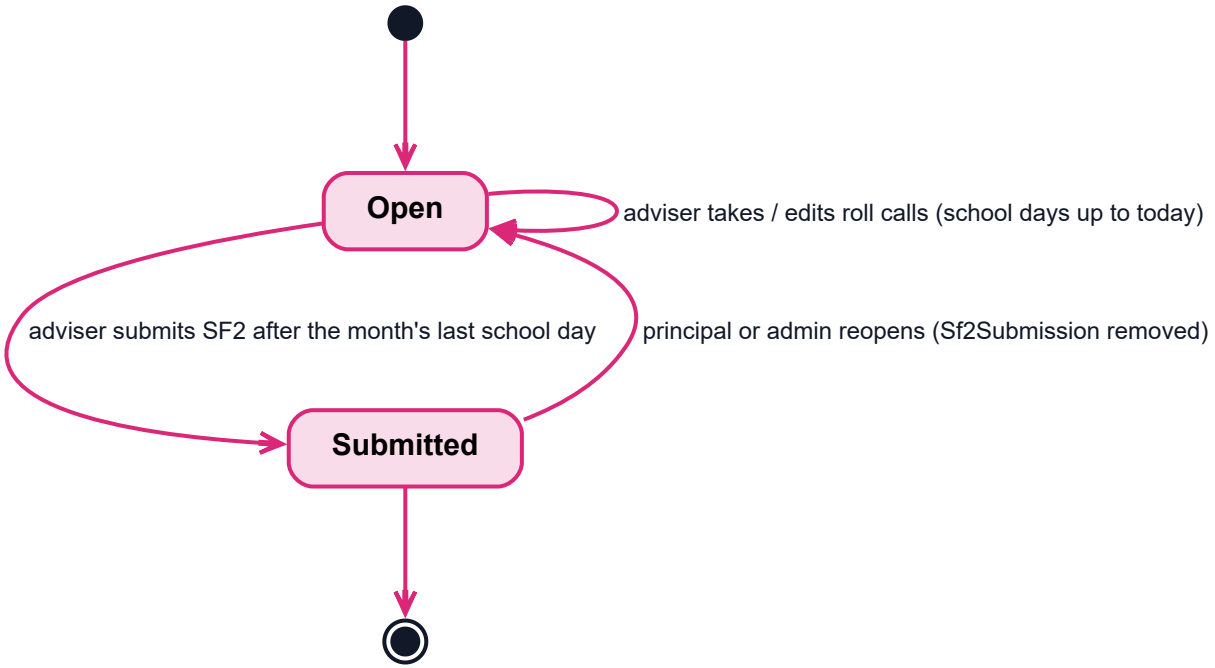
A standard use case diagram: the actors on the left, each use case as an ellipse inside the system frame, and a line for each use case an actor starts. A dashed line has a note on when it applies.



Adviser · Principal / Administrator — Line colour = the actor who starts it.

Attendance — Ellipse colour = the module the use case belongs to.

3.1 Flow: A month of attendance



Attendance Drawn in the colour of the module this flow belongs to.

4. Use cases (changes)

Each card is one use case command sent through the use case bus. It runs in one database transaction: if any rule refuses, nothing is saved and the message in *Alternatives and refusals* is shown to the user.

ATT-01 Take the day's roll call

Primary actor	Adviser
Goal	Record who was absent, late or cutting classes on a school day.
Where (route)	`teacher_class_attendance /teacher/class/attendance/{date}`
Preconditions	• The teacher advises a section this school year; the date is a school day up to today; the month is not submitted.
Main flow	1. The adviser opens the roll call (today by default) and sees the learners who count that day. 2. Marks each exception: Absent, Late or Cutting; everyone else is present. 3. Saves. The system takes the section lock, ignores marks for learners who don't count that day, creates or updates the roll call and replaces its marks. 4. The activity log records "Took attendance of ..." or "Updated attendance of ...".
Alternatives and refusals	• "That date is outside the school year." • "No classes on weekends." • "No classes: <day off reason>." • "You can't take attendance for a future date." • "<Month Year> is submitted and locked." • "You do not advise a section this school year." / "You can no longer take attendance for this section."
Result (data changed)	RollCall (one per section and day) and its AttendanceMark rows.
In the code	`TakeRollCallCommand → AttendanceRecorder::takeRollCall`

ATT-02 Submit the month's SF2

Primary actor	Adviser
Goal	Hand the month's attendance to the principal.
Where (route)	`teacher_class_attendance_submit` `/teacher/class/attendance/month/{month}/submit`
Preconditions	• The month's last school day (or the school year) has passed; not yet submitted.
Main flow	1. The adviser opens the month page, which shows the SF2 figures and any school days without a roll call. 2. If days are missing, the adviser ticks the confirmation that everyone was present on them. 3. Submits. The month is locked and "Submitted SF2 of <section> for <month>" is logged.
Alternatives and refusals	• "<Month Year> is already submitted." • "You can submit <Month> once its last school day has passed." • "N school days have no attendance. Confirm that everyone was present on those days."
Result (data changed)	Sf2Submission (section, month).
In the code	`SubmitSf2Command → AttendanceRecorder::submit`

ATT-03 Reopen a submitted month

Primary actor	Principal, Administrator
Goal	Let the adviser correct a month that was already submitted.
Where (route)	`school_attendance_reopen` `/schools/{id}/sections/{section}/attendance/{month}/reopen`
Preconditions	• Manages the school; the month is submitted.
Main flow	1. From the section's month page the principal presses Reopen. 2. The submission is removed and "Reopened SF2 ..." logged.
Alternatives and refusals	• Not allowed by AttendancePolicy (403).
Result (data changed)	Sf2Submission removed.
In the code	`ReopenSf2Command`

5. Views, downloads and reports (read only)

Each row is one query sent through the query bus. Queries change nothing.

Page / file	Who	Route	What it shows	Query
Roll call page	Adviser	`teacher_class_attendance`	Who counts on the day, marks so far, and whether it can be taken.	`GetRollCallPageCommand`
Month page	Adviser, Principal	`teacher_class_attendance_month; school_attendance_section`	The month with SF2 figures, missing days, lock state.	`GetAttendanceMonthCommand`
School attendance overview	Principal, Administrator	`school_attendance /schools/{id}/attendance (principal_attendance)`	Today's attendance and every section's month, with follow-ups.	`GetAttendanceOverviewCommand`
SF2 file (.xlsx)	Adviser, Principal	`SF2 download links on the month pages (teacher_class_sf2)`	DepEd SF2 filled into sf2-template.xlsx.	`GetSf2FileCommand`